

JavaScript

Candidates for this exam should be able to recognize and write syntactically correct JavaScript code that will logically solve a given problem and use data types supported by JavaScript.

Candidates are expected to have at least 150 hours of instruction or hands-on experience with the JavaScript programming language. Candidates should be familiar with JavaScript features and capabilities, and understand how to write, debug, and maintain well-formed, well-documented JavaScript code.

To be successful on the test, the candidate is also expected to have the following prerequisite knowledge and skills:

- 8th grade reading skills
- Algebra I
- Fundamental knowledge of HTML
- Fundamental knowledge of CSS

1. JavaScript Operators, Methods, and Keywords

1.1 Complete and debug code that uses assignment and arithmetic operators

- Assignment, increment, decrement, addition, subtraction, division, multiplication, modulus, compound assignment operators ($+=$, $-=$, $*=$, $/=$, $\%=$)

1.2 Apply JavaScript best practices

- Comments, indentation, naming conventions, noscript, constants, reserved keywords, debugger keyword, setting breakpoints, console.log

1.3 Evaluate the use of internal and external scripts

- When to use, how to use, and what happens when scripts are used at multiple levels

1.4 Implement exception handling

- try, catch, finally

1.5 Complete and debug code that interacts with the Browser Object Model (BOM)

- Displaying dialogs, determining screen size

2. Variables, Data Types, and Functions

2.1 Declare and use variables of primitive data types

- Number, Boolean, String, null, undefined, type of operator, type-checking functions, use strict, converting between data types (parseInt, parseFloat), formatting numbers, string operations, eval(), toFixed(), toLocaleString(), toPrecision(), single quote vs. double quote (nesting), initialization

2.2 Declare and use arrays

- Single-dimensional arrays; multi-dimensional arrays; iteration; initialization; defining, sorting, and searching an array; push, pop, shift, and unshift methods; length property; accessing an array element



IT SPECIALIST EXAM OBJECTIVES

2.3 Complete and debug code that uses objects

- Properties, methods, instantiation, Date object, retrieving date and time parts, localizing date format (MM/DD vs DD/MM), adding and subtracting dates

2.4 Complete and debug code that uses built-in Math functions

- random, round, abs, floor, ceil, min, max, pow, sqrt

2.5 Complete and debug functions that accept parameters and return values

- Reusable code, local vs. global scope, redefining variables, passing parameters, value vs. reference, return values

3. Decisions and Loops

3.1 Evaluate expressions that use logical and comparison operators

- !=, <, >, <=, >=, !, ==, &&, ||

3.2 Complete and debug decision statements

- Single alternative (if), dual alternative (if else), multiple alternative (switch), nested if

3.3 Complete and debug loops

- for, for in, while, do while, break, continue

4. Document Object Model

4.1 Identify and construct the Document Object Model (DOM) tree

- window, document, body, other HTML elements

4.2 Identify and handle document, form, keyboard, and mouse events

- onload, onfocus, onblur, onchange, onkeydown, onkeyup, onkeypress, onclick, onmouseover, onmouseout

4.3 Complete and debug code that outputs to an HTML document

- document.write, innerHTML, textContent

4.4 Complete and debug code that locates, modifies, and adds HTML elements and attributes to documents

- getElementById, getElementsByTagName, getElementsByClassName, setAttribute, createElement

4.5 Create events using event handlers and listeners

- DOM events, HTML attribute event, addEventListener

5. HTML Forms

5.1 Complete and debug code that retrieves form input and sets form field values

- Retrieving form values; identifying the DOM path; getting values from different types of elements; prepopulating, masking, and updating values

5.2 Complete and debug code that performs input validation

- Case, string comparisons, Not-A-Number (NaN), not blank

5.3 Describe the form submission process

- onsubmit, POST vs. GET, potential targets for submission

